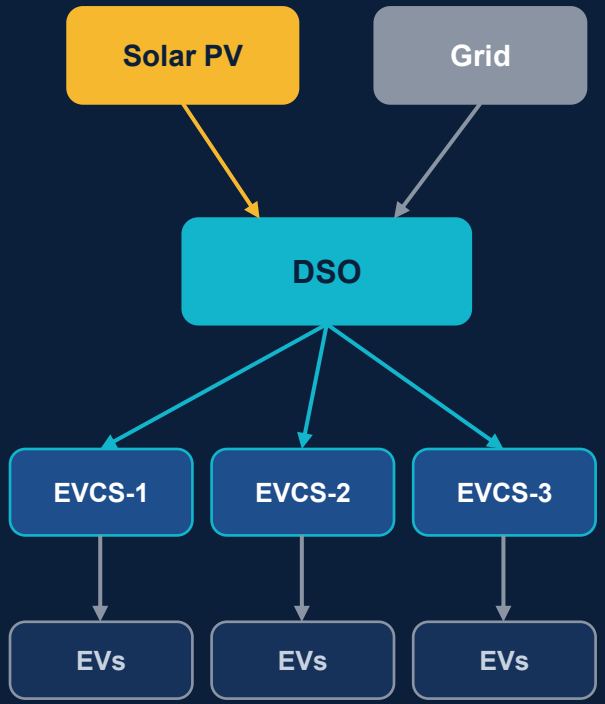
B.TECH PROJECT MID-TERM PRESENTATION

Hierarchical Energy Management and MILP-Based EV Charging Scheduling for a Solar–Grid Integrated EV Charging Network

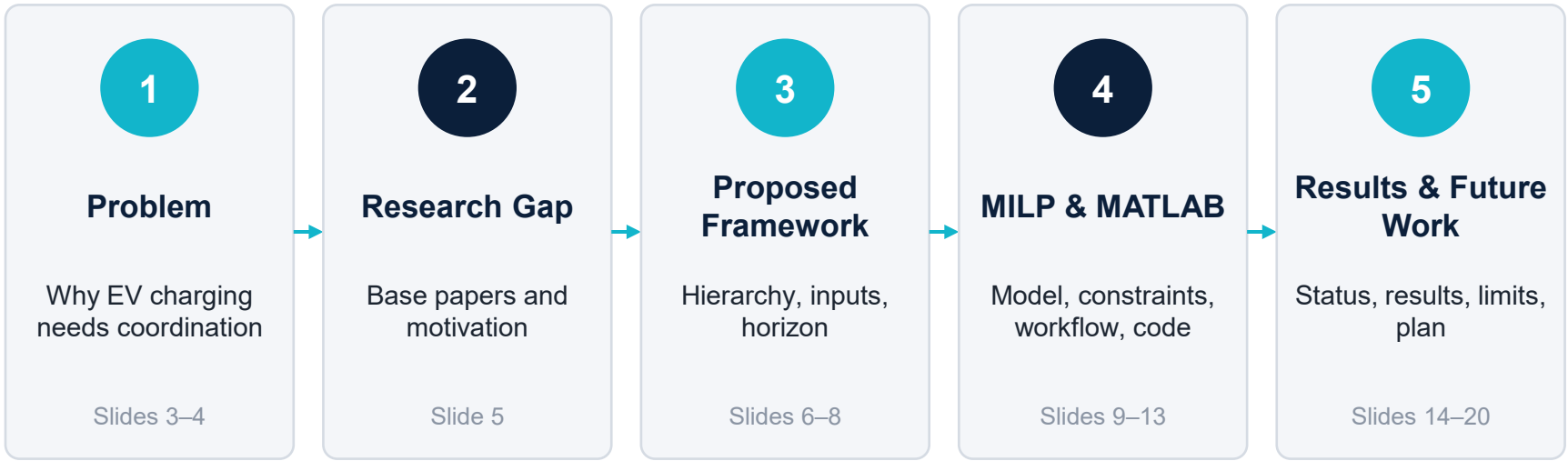
Group 24
Shreya Dubey (20235082)
Harshita Singh Bagri (20235123)

Supervisor: Dr. Prashant Kumar Tiwari
Department of Electrical Engineering
MNNIT Allahabad, Prayagraj | 2026



Conceptual schematic

Roadmap: from the problem to preliminary evidence



Why coordinate EV charging with solar and grid supply?

EV demand is user-driven

Arrival time, stay duration and required SoC differ for every vehicle.

Solar is time-varying

PV output may not coincide with EV arrival and departure windows.

Grid remains necessary

Grid covers periods of insufficient solar generation.

Uncoordinated charging

Simultaneous demand may raise grid dependency and under-use solar.

Scheduling flexibility lets part of the charging be shifted toward solar hours. Source: project report §1.1, §1.4.

Problem: schedule EVs across stations and ports under constraints

CHALLENGE

- Allocate solar and grid energy among multiple EVCSs and ports
- Respect arrival/departure windows and port availability
- Respect station, port, EV, grid and solar power limits
- Deliver the required SoC before departure

OBJECTIVE

Hierarchical framework

DSO allocation + EVCS scheduling

15-min MILP model

Multiple stations and ports; SoC modelling

Economic objective

System, participant and solar terms

MATLAB + intlinprog

Readable EV-by-EV 15-min schedule

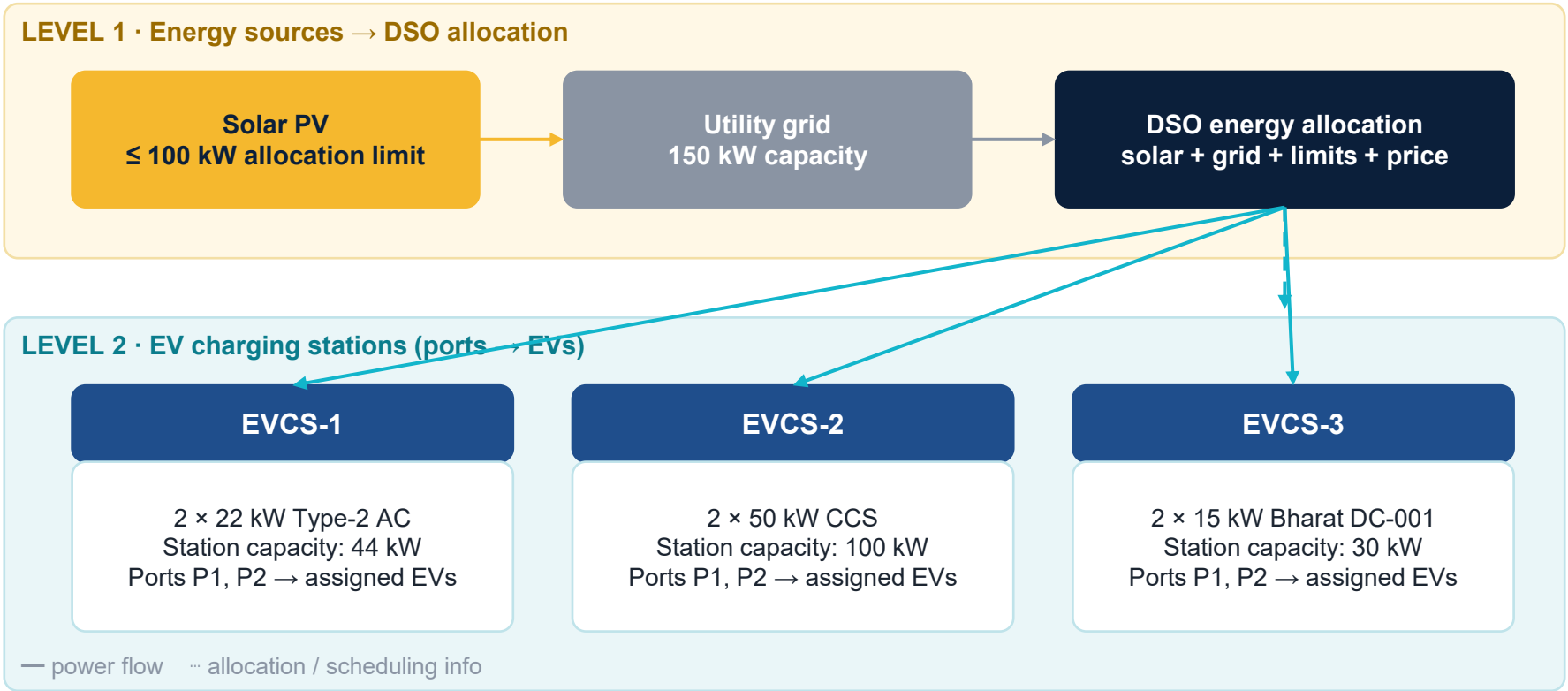
Scope: deterministic, offline day-ahead scheduling over 96 slots. Real-time communication and hardware validation are outside this stage.

Literature Review

Aspect	[1] Utility Energy Optimization for Residential Microgrid With Electric Vehicle Loads — IEEE Access, 2026	[2] Energy Management in Microgrids Using Model-Free Deep Reinforcement Learning Approach — IEEE Access, 2025
Method	Rule-based MPMC with optimization (fmincon, quadprog)	Model-free DRL (DDPG), MDP formulation
System	Residential microgrid: PV, ESS, EV, household load, grid	Grid-tied microgrid, IEEE 33-bus, EV charging stations
Resolution	Hourly updating (MPMC)	Hourly EMS decisions
Relevance	Renewable/grid/EV coordination structure	Motivation for advanced EVCS energy management

Research gap (as stated in report): explicit integration of DSO-level allocation, multiple EVCSs and ports, EV-specific windows and SoC needs, solar/grid allocation, capacity constraints and a multi-term economic objective. MILP supplies binary EV-to-port decisions and continuous power decisions.

Proposed architecture: two-level hierarchy



Inputs the scheduler needs

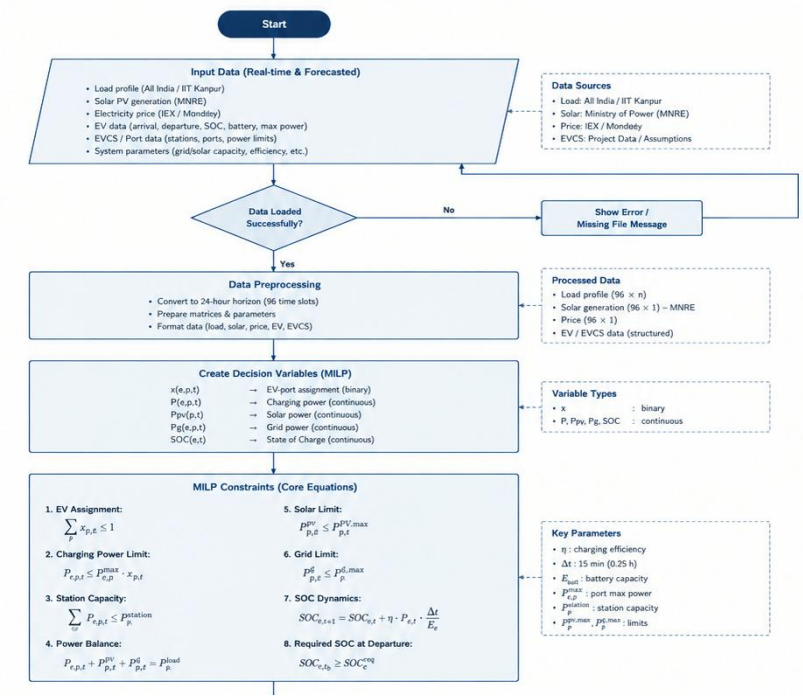
EV	Arrive (slot)	Depart (slot)	$SoC_0 \rightarrow$ req. (%)	Battery (kWh)	Pmax (kW)
EV1	20	48	30 → 80	40	7
EV2	28	60	45 → 90	60	11
EV3	12	40	25 → 80	50	7
EV4	40	72	50 → 90	75	11
EV5	52	84	20 → 80	40	7

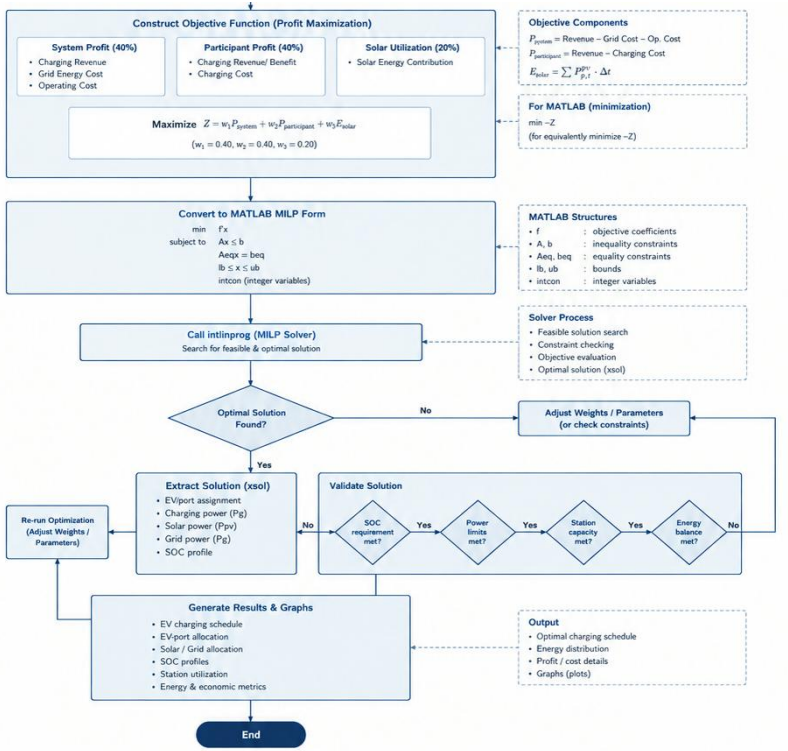
System parameters
$\Delta t = 0.25$ h, $T = 96$
Efficiency $\eta = 0.95$
Grid 150 kW · Solar limit 100 kW
Tariff 10 INR/kWh (assumed)
Operating cost 1 INR/kWh (assumed)
Weights 0.40 / 0.40 / 0.20

Also required: port availability, port power ratings (15/22/50 kW), station capacity, hourly solar and grid availability, and a 96-slot price input.



System overview: end-to-end MILP scheduling flowchart





Part 1: inputs → preprocessing → variables → constraints

Part 2: objective → solver → validation → results

MILP formulation: assignment decides where, power decides how much

Sets & variables

$e \in E$ EVs · $s \in S$ stations · $p \in P_s$ ports · $t \in T$ slots

$x(e,s,p,t) \in \{0,1\}$ EV–port assignment
$P(e,s,p,t) \geq 0$ charging power
$P_PV(e,s,p,t) \geq 0$ solar power
$P_G(e,s,p,t) \geq 0$ grid power
$0 \leq SOC(e,t) \leq 1$ state of charge

Objective

$$\max Z = w_1 \Pi_sys(norm) + w_2 \Pi_participant(norm) + w_3 E_solar(norm)$$

$w_1 = 0.40, w_2 = 0.40, w_3 = 0.20$
Implemented as $\min(-Z)$

Central constraints

$P = P_PV + P_G$
$SOC(e,t+1) = SOC(e,t) + \eta \cdot \Sigma\Sigma P \cdot \Delta t / E_e$
$SOC(e,D_e) \geq SOC_e_req$

Constraints: power limits and EV energy requirements

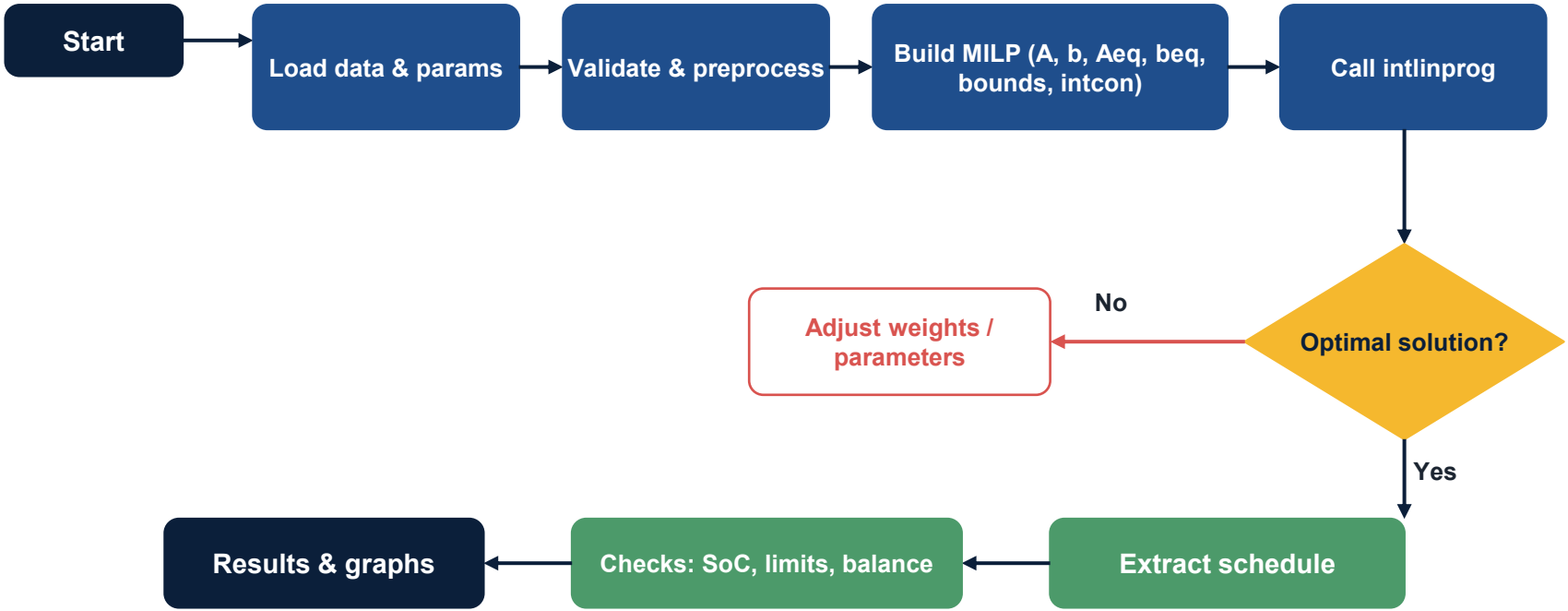
Assignment & power

- One port per EV per slot (4.6)
- Port availability and single occupancy (4.7–4.8)
- Port rating: $P \le P_{max_port} \cdot x$ (4.9)
- EV limit $\Sigma P \le P_{max_EV}$ (4.10)
- Station capacity (4.11)
- Grid and solar limits (4.12–4.13)
- Power balance: $P = P_{PV} + P_G$ (4.14)

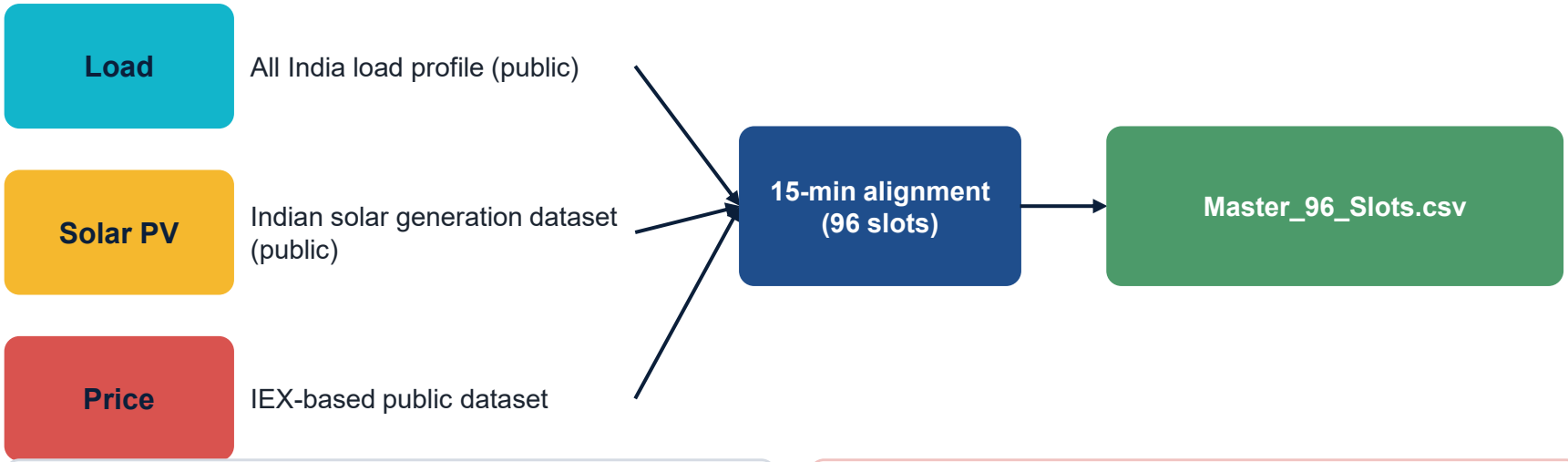
EV energy & timing

- Initial SoC fixed (4.15)
- SoC dynamics with $\eta = 0.95$ (4.16)
- Required SoC at departure (4.17)
- SoC bounds 0–1 (4.18)
- Charging only within connection window (4.19–4.20)

Offline optimization workflow in MATLAB



Data: public datasets combined with project-configured inputs



Manually configured
EV arrival/departure, battery and SoC; EVCS ports (BEE specs + project configuration); grid/solar limits

Study assumptions
Tariff, operating cost, objective weights. Not measured field data.

MATLAB implementation: modular pipeline



Directory structure

```
01_Data  02_Preprocessing
03_Mathematical_Model
04_MATLAB (Data_Loading,
   Optimization, Results)
05_Simulation  06_Results
07_Figures  08_Documentation
```

Decision vector (12,000 variables)

Binary assignment x · 2,880
Charging power P · 2,880
Solar P_PV · 2,880
Grid P_G · 2,880
SoC · 480

Work completed so far, and what is still open

COMPLETED

- Project data structure
- Load, solar, price preprocessing
- Master 96-slot dataset
- EV input data, EVCS configuration, system parameters
- MILP formulation and MATLAB solver implemented
- Schedule extraction implemented
- Preliminary feasibility testing

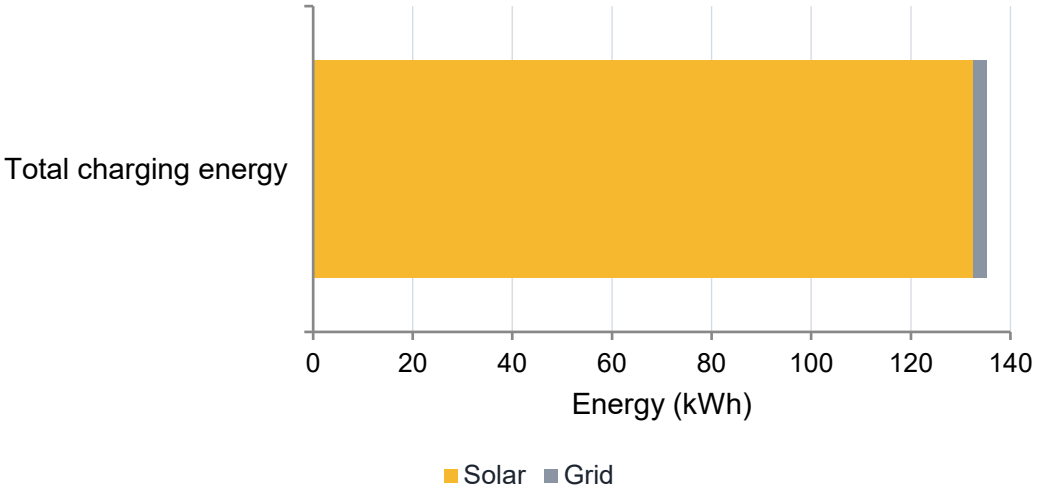
IN PROGRESS / REMAINING

- Ongoing: station-selection refinement
- Ongoing: economic differentiation of EVCSs
- Remaining: comparative validation
- Remaining: final performance analysis

Preliminary run: optimal solution, 98.01% solar energy

5	6	96	12,000	2,880
EVs	ports	time slots	variables	integer (pre-presolve)

Solver status: optimal, zero reported optimality gap. Result specific to the present inputs and constraints.



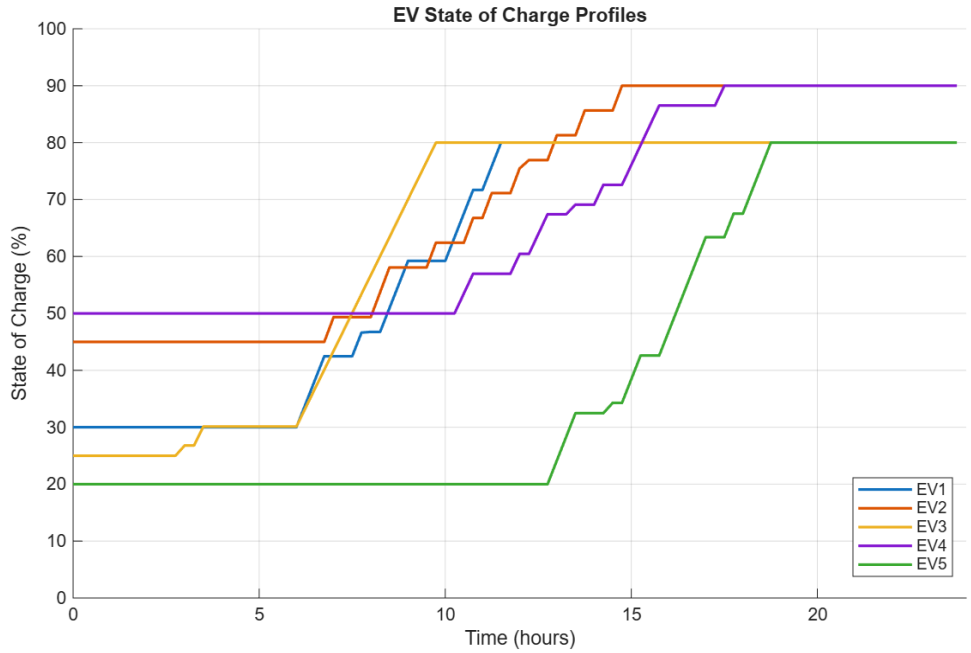
135.2632 kWh
Total charging energy

132.5658 kWh
Solar energy

2.6974 kWh
Grid energy

98.01%
Solar share (solar / total)

EV SoC profiles: all five EVs reach their required SoC



MATLAB output: EV state of charge profiles

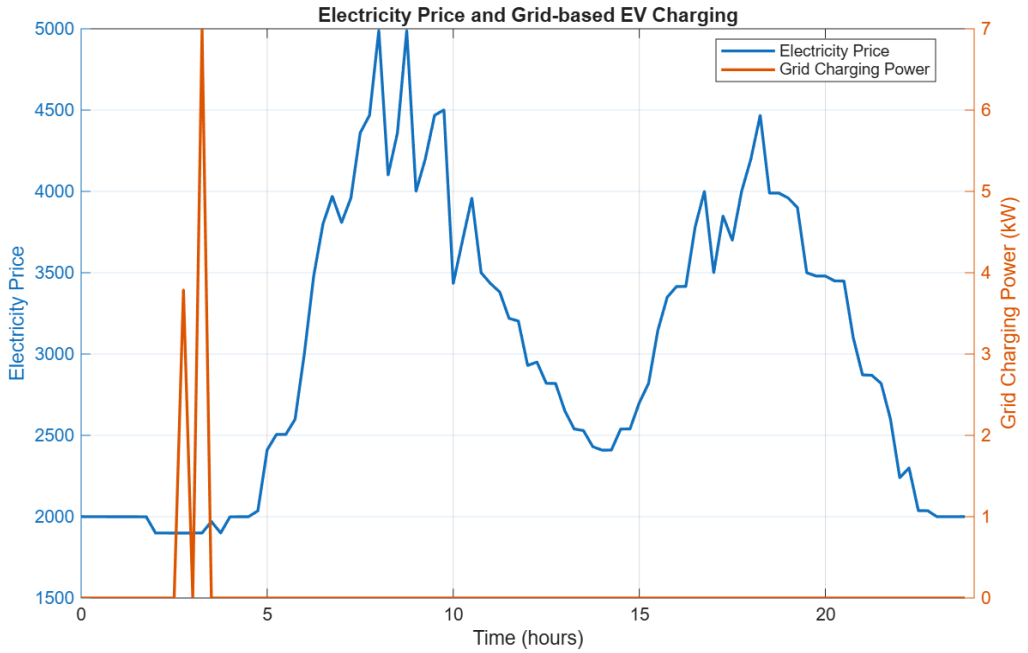
EV	Initial	Req.	Final
EV1	30	80	80
EV2	45	90	90
EV3	25	80	80
EV4	50	90	90
EV5	20	80	80

SoC in %

Validates: required SoC met, 5/5 EVs

Does not validate: economic objective or practical operation

Grid charging occurred only in a brief low-price window



MATLAB output: price input and grid charging power

Observation
Grid power is non-zero only around 2.5–3.5 h; it is zero for the rest of the day.

Total grid energy in this run: 2.6974 kWh (about 1.1% of charging energy).

Caution: price units are unverified

Critical observations: feasible, but not yet final

Zero-power port assignments	x can be 1 with P = 0, so entries read “connected, not charging”. Planned: connection vs charging state.
Port switching across slots	EV may change ports mid-session. Planned: fixed station/port variable y(e,s,p) or switching penalty.
No station-specific economics	Fixed tariff makes participant cost proportional to energy; stations are not differentiated.
Price units unverified	Price interpretation must be checked before monetary conclusions.
No baseline comparison yet	Deterministic model; comparison cases not completed.

Remaining work: refine, compare, test sensitivity



Refine model

Fixed station/port selection; connected vs charging state

Station economics

Station-specific tariffs, operating costs, switching penalties; verify price units

Comparative cases

Uncontrolled, grid-only, solar-priority, MILP without and with station economics

Sensitivity analysis

Solar, grid capacity, windows, EV count, tariff, weights, efficiency

Final evaluation

Metrics: solar use, grid energy, peak demand, SoC, profit, run time

Conclusion

Problem	Coordinate EV charging across stations and ports using solar and grid energy.
Implemented	Hierarchical MILP in MATLAB; feasible 15-min schedule meets required SoC for 5 EVs.
Next	Comparative validation, station-aware economics and sensitivity analysis.

Thank You

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Supervisor: Dr. Prashant Kumar Tiwari